Kürten 2021 HNSCC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

9/20 samples (≥182 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached); the low-cycling embryonic lobe (5,955 cells) carries the trajectory.

Gate 2 — module geography → GO (non-branching)

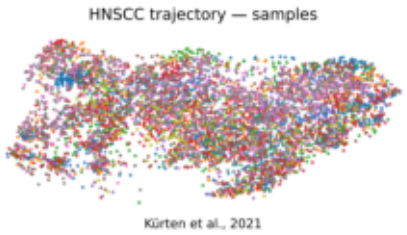
Gast/pSom co-vary (r=+0.33) and decline along pseudotime (ρ=−0.32 / −0.38); Crest anti-correlated (r=−0.77 / −0.64), rising to a terminal readout (ρ=+0.44; crest-high decile mean pt=0.99).

Gate 3 — cross-repression → NA

Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

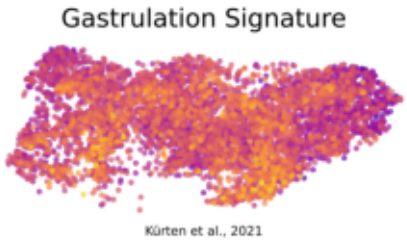
Manifold — samples



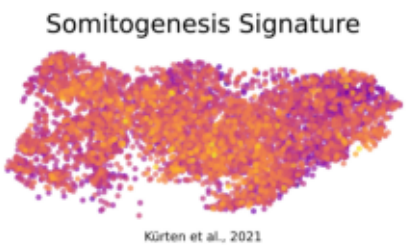
Pseudotime



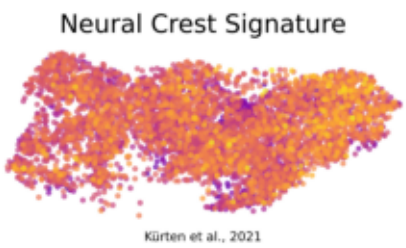
Gastrulation



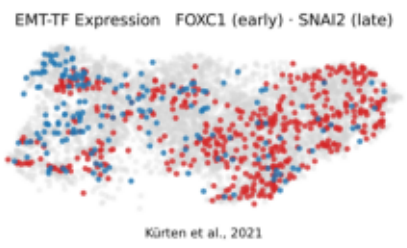
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

Head & Neck SCC (Kürten et al., 2021), 10x — 33 patients / 62,700 cells (master scope); malignant only: 11,122 across 9 samples (≥182) enter preprocessing; 5,955 embryonic-lobe cells enter the trajectory. Non-branching Linear-pSom/Crest.

Verdict.

Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Kürten's malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. EMT-TFs FOXC1 (early/root) → SNAI2 (late/terminus); canonical SNAI1/ZEB1 near the detection floor (2.0%/2.3%).

Parameter choices

Min cells/sample	≥182 (keeps 9/20; gap 89→376)	**Lobe selection**	embryonic (low-cycling 0.22) vs 0.46
Root node	tip 0 = Gast/pSom-high, Crest-low	**Integration**	Harmony(sample) θ=10; PCA 50PC; UMAP spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100	**Compartments**	Embryonic 5,955 / Cycling 5,167
Geometry	Linear (non-branching); no hub	**EMT-TFs**	FOXC1 (early/root) → SNAI2 (late/terminus); SNAI1/ZEB1 near floor